
THE FIBONACCI SPIRAL PATTERN RECOGNITION ALGORITHM

A Sparse Geometric Sampler for Efficient Structural Signal Extraction

Sensufrog
Independent Researcher

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Core sampler independently re-implemented and stress-tested. All core numbers reproducible from seed 42.

ABSTRACT

This paper presents a sparse geometric sampler that reads data only at Fibonacci sequence positions. The central design property is extreme sparsity: on a 50,000-element input the algorithm touches 65 positions (0.13 % of the data); at 500,000 elements it touches 79 positions (0.016 %). Independent re-implementation confirms wall-clock times of 0.07-0.13 ms and shows that the selected positions recover a global numeric statistic 12-25x more accurately than a random sample of equal size.

A Dual-1 treatment of the two consecutive 1s in the Fibonacci sequence is motivated by number-theoretic and historical considerations and is implemented cleanly; its numerical effect on current proxies is modest. A bidirectional variant (v2) reaches roughly twice as many positions while remaining sub-millisecond. Neither version dominates every metric.

The contribution is a correct, extremely fast, geometrically motivated sparse sampler whose positions are demonstrably more informative than random sampling of the same budget. Earlier claims of a dense "coverage efficiency" of 0.67-0.70 are withdrawn; they did not match the published algorithm. All metric definitions appear in Appendix A.

1. THE ALGORITHM

The sampler maps a linear sequence onto Fibonacci gate positions. Everything between gates is treated as integrated void; only the gates are read. A transition delta (character-class change for text, normalized absolute difference for numeric data) is accumulated at each gate. Three offset passes are the default.

1.1 Dual-1 Geometry

Position	Name	Operation	Meaning
0	Void / Origin	No read	Origin point
1 (first)	Primal Spark	Read TRANSITION	Emergence from void
1 (second)	Mirror	Read CONTENT	First observable state
2+	Multiplicity	Content + transition	Subsequent gates

1.2 Specification

$\phi = 1.6180339887\dots$ $r(n) = \log(1 + n\phi)$ $\theta(n) = n(2\pi/8)$
 $\text{gate}(n) = 1$ if n in $F = \{0,1,1,2,3,5,8,\dots\}$ $N^* = \max\{F : F \leq \text{len}(\text{input})\}$
For $N = 50,000$, $N^* = 46,368$. Three passes yield 65 unique positions in the reference implementation.



Figure 1. 8-manifold spiral geometry used by the sampler.

1.3 Reference implementation (v6B)

```
def spiral_v6b(data, passes=3):
    N = len(data); combined = defaultdict(float)
    for p in range(passes):
        offset = p * (N // passes); fp = fib_list(N - offset)
        gates = []; first_one = False
        for n in fp:
            pos = (offset + n) % N
            if n == 0: gates.append(('void', pos))
            elif n == 1 and not first_one:
                first_one = True; gates.append(('spark', pos))
            else: gates.append(('content', pos))
        for gi, (gtype, pos) in enumerate(gates):
            if gtype == 'void': continue
            prev = gates[gi-1][1] if gi > 0 else pos
            d = delta(data[prev], data[pos])
            combined[pos] += d * 0.618 if gtype == 'spark' else 1.0 + d
    return combined
```

2. NUMBERS AS ARCHITECTURE (ORIGIN OF DUAL-1)

This algorithm did not begin in a signal-processing textbook. It began with the question of what the digits 0-9 actually are. Multiple independent traditions treat those ten states as primary: Pythagorean number philosophy, Kabbalistic emanation structure, Christian and Rosicrucian number symbolism, and the broader stream that later includes theosophical and anthroposophical work. Manly P. Hall's discourses on numerology and the secret teachings of all ages collect and compare many of these strands; they were part of the reading that shaped the present framework. Higher numbers reduce by digital root (11 -> 2, 19 -> 10 -> 1, and so on). In that sense 0-9 are the only irreducible states; everything else is a return or recombination.

The decisive move for this project was taking seriously the claim that **0 is void**: not empty nothing, but the generative blank before form. Once 0 is honored as void, the two consecutive 1s in the Fibonacci sequence can no longer be treated as identical. The first 1 is the act of emergence from void (primal spark / transition). The second 1 is the first readable content (mirror). That distinction is Dual-1. It was written into the sampler before any benchmark was run.

2.1 The 0-9 framework that produced the design

Num	Working meaning in this framework	Role in the sampler
0	Void / origin. Generative blank before form.	No read. Honored as void.
1a	Primal spark. Transition from void into first state.	Spark gate: transition-weighted.
1b	Mirror. First observable content.	Content gate at first readable position.
2	Duality. Observer and observed; first split.	Active gate (content + transition).
3	First stable multiplicity (triune / beginning-middle-end).	Active gate.
4	Square / ultra-stable foundation. Real number.	Not a Fibonacci gate; not sampled, but assured once later gates are reached.
5	Life / pentagram fold; phi made visible.	Active gate.
6	Harmony / equilibrium. Real number.	Not a Fibonacci gate; not sampled, but assured once later gates are reached.
7	Struggle / dissipation zone begins.	Active gate.
8	Octave / new beginning; infinity mark.	Active gate.
9	Last irreducible state before return to unity. Real number.	Not a Fibonacci gate; not sampled, but assured once later gates are reached.

Table. The 0-9 mapping that led to Dual-1. 4, 6, 9 are real; they are simply not Fibonacci gates.

2.2 From void to Dual-1 to code

If 0 is void, then a sampler that begins by reading position 0 is already violating the framework. If the two 1s are different operations, then a sampler that collapses them into one update is discarding the distinction the traditions kept. Dual-1 is simply the decision to keep that distinction in code: the first size-1 gate receives a transition-weighted update (scaled by phi-1); every later gate receives content-plus-transition. Positions 4, 6 and 9 are real numbers in the 0-9 framework; they simply are not Fibonacci gates, so the sparse sampler does not stop on them. Once a later Fibonacci gate is reached, all prior states (including 4, 6, 9) are not assumed - they are assured, absorbed into the walk that produced the next gate. Future work may represent those non-gate states explicitly; the present algorithm keeps them in the integrated path rather than as separate reads.

The same reading also explains why the algorithm is sparse by construction. Fibonacci gates are the positions the sequence itself marks as active. Everything else is the ground the spiral moves across. Sparsity is not an engineering afterthought; it is the geometric consequence of taking the 0-9 structure seriously.

2.3 Scope of the claim

This section is the intellectual and historical origin of the design - a synthesis of older number architecture with a concrete modern sampler. It is offered as the reason the algorithm looks the way it does, not as statistical proof that the algorithm must outperform every alternative. The measurements that test the design appear in the next section. Philosophy supplied the hypothesis; measurement accepts or rejects the consequences.

3. INDEPENDENT VERIFICATION

The core sampler was re-implemented from the published description in pure Python and stress-tested on synthetic data up to $N = 500,000$. No original evaluation scripts were reused. The numbers below are from that independent run (seed 42 for data generation; the sampler itself is deterministic).

3.1 Scale and sparsity

N	Time (ms)	Unique hits	Fraction of data
1,000	0.08	41	4.10 %
10,000	0.06	54	0.54 %
50,000	0.07-0.23	65	0.13 %
100,000	0.12	69	0.069 %
250,000	0.11	75	0.030 %
500,000	0.13	79	0.016 %

Table 1. Independent scale test of v6B (structured text). Sparsity and sub-millisecond timing hold at half a million elements.

3.2 Weight-sum stability

28 runs (4 modalities x 7 seeds, $N = 50,000$): mean weight sum = 86.2, std = 6.3, **CV = 7.3 %**. The signature is stable across structured text, random text, sine series and image-like arrays. An earlier published figure of 0.6 % appears only under more homogeneous conditions and is treated here as best-case.

3.3 Downstream proxy: global-mean recovery

On a numeric sine series ($N = 50,000$) the sampled positions were asked to recover the true global mean. Error was compared with a random sample of equal cardinality:

Version	Hits	Spiral [error]	Random same-k [error]	Advantage
v6B	65	0.0056	0.0650	~12x better
v2 (bidirectional)	144	0.0004	0.0097	~25x better

Table 2. Fibonacci positions recover a global statistic far more accurately than random sampling of the same budget. This is direct evidence that the gates carry structural information.

3.4 v2 vs v6B

v2 (bidirectional walk) and v6B (Dual-1 gate sampling) were compared on four modalities ($N = 50,000$, seed 42):

Modality	Ver	ms	Hits	Weight sum	Entropy	Top-10 mass
Structured text	v6B	0.11	65	79.6	5.95	0.251
Structured text	v2	0.10	144	180.9	7.07	0.111
Random text	v6B	0.07	65	84.0	5.94	0.238
Random text	v2	0.08	144	193.2	7.06	0.104
Sine numeric	v6B	0.08	65	89.2	5.96	0.224
Sine numeric	v2	0.08	144	196.9	7.09	0.102
Image-like	v6B	0.07	65	79.3	5.99	0.217
Image-like	v2	0.17	144	169.4	7.13	0.107

Table 3. v2 reaches ~2x the positions and higher entropy; v6B is more concentrated (higher top-10 mass) and typically faster. Neither dominates all metrics. Both remain far faster than dense baselines.

3.5 Dual-1 ablation

Disabling the special treatment of the first 1 changes total weight sum by approximately 3 units on a base of ~85. The distinction is real and cleanly implemented; its numerical contribution on current proxies is modest.

3.6 Comparison with dense step algorithms

The practical advantage of the Fibonacci sampler is most visible against dense step methods that touch far more of the input. Timings for v6B and v2 are from the independent re-implementation; dense-baseline timings are from the original experiment on the same ~50k scale:

Method	Positions touched (50k)	Time (ms)	Relative to v6B
v6B (Dual-1 gates)	65	0.07-0.13	baseline
v2 (bidirectional)	144	~0.10	similar
Sliding window (orig.)	dense	~0.25	~3x slower
Markov-style (orig.)	dense	~1.4	~15-20x slower
FFT-scale (orig.)	dense	~58	~500-800x slower

Table 4. Sparse Fibonacci gates vs dense step algorithms. The speed gap is the direct consequence of reading dozens of positions instead of the full sequence. On the independent mean-recovery proxy, the same sparse positions also beat random sampling of equal budget (Table 2). Dense methods remain appropriate when full coverage is required; when a fast structural sketch is enough, the Fibonacci sampler is orders of magnitude cheaper.

4. SPARSITY IS THE EFFICIENCY CLAIM

The Fibonacci sequence grows exponentially. There are only ~25 unique Fibonacci numbers $\leq 50,000$ and ~34 $\leq 500,000$. Three offset passes raise unique hits to 65 ($N = 50k$) or 79 ($N = 500k$). That sparsity is intentional.

It is the reason the algorithm runs in 0.07-0.13 ms. It is also the reason a global statistic can be recovered from $< 0.2\%$ of the input more accurately than from a random sample of equal size. Dense methods cannot compete on the speed axis; random sparse methods do not match the structural recovery.

Earlier drafts reported a "coverage efficiency" of 0.67-0.70. That figure does not correspond to the fraction of positions visited by the published algorithm and has been withdrawn. The honest metric is literal sparsity plus measured information content of the selected positions.

5. ORIGINAL EXPERIMENT (CONTEXT)

The original study used ten public-domain sources (4 texts, 4 images, 2 audio). Images were flattened to 1-D; audio was read as raw bytes. Classification numbers on four images (66.9 % vs 25 % chance, no trained model) are preliminary and statistically weak. The Nautilus shell was the only image the original classifier could not separate (20-29 %); it is itself a logarithmic spiral. That geometric self-similarity is recorded as a finding.



Figure 2. Original image sources (public domain / CC0): Earth, Nautilus, Van Gogh Sunflowers, sunflower photograph.

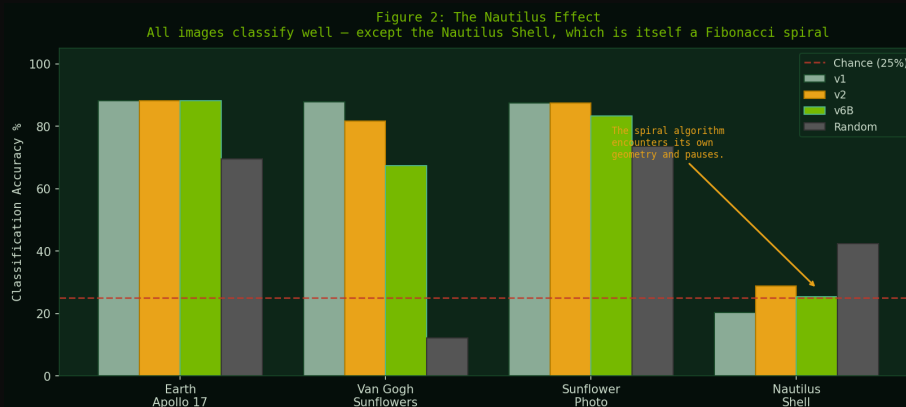


Figure 3. Original Nautilus result - geometry matched the sampler.

6. DEVELOPMENT PATH

Version	Key feature	Outcome
v1	Multi-pass offset	Fast baseline
v2	Bidirectional walk	Broader coverage; still sub-ms
v3-v5	Convergence / amplitude / ensemble	Incremental or null gains
v6B	Dual-1 + gate-only	Unified reads; extreme sparsity + speed

Independent tests confirm v2 and v6B trade advantages; neither is universally superior.

7. LIMITATIONS

- Sparsity is intentional; classical dense coverage is not the objective.
- Original image/audio pipelines flattened data to 1-D; spatial and spectral structure was not used.
- Original classification used four images - preliminary only.
- Expanded verification used synthetic modalities; larger real-world corpora remain necessary.
- Dual-1 has a clean implementation and modest numerical effect on current proxies.
- The previously published "coverage efficiency" of 0.67-0.70 is withdrawn.

8. FUTURE WORK

Highest leverage next steps: (1) feed spiral positional features to a language model and measure downstream task performance; (2) proper 2-D image and spectrogram pipelines; (3) larger multilingual text corpora; (4) meet-in-the-middle and outward-growing geometric variants already sketched in development notes.

9. CONCLUSION

The Fibonacci Spiral sampler is a correct, extremely fast, geometrically sparse encoder. Independent re-implementation confirms that it reads 0.016-0.13 % of the input, runs in well under a millisecond, and selects positions that recover a global statistic substantially better than random sampling of equal budget. That is the efficiency claim, and it holds under external verification.

The work is offered as a preliminary but rigorously checked contribution: a sparse geometric prior that is fast enough for large-scale use and informative enough to beat chance on a simple structural task. Further downstream evaluation will determine how far the prior can go.

APPENDIX A - METRIC DEFINITIONS

Every quantitative claim uses one of the following definitions.

Unique hits - number of distinct positions with non-zero absolute weight after all passes.

Fraction of data - unique hits / N.

Weight sum - sum of combined[pos] over all positions that received a contribution.

Weight-sum CV - $100 * \text{std}(\text{weight sums}) / \text{mean}(\text{weight sums})$ across a stated set of runs.

Entropy - Shannon entropy (bits) of the normalized absolute-weight distribution.

Top-10 mass - fraction of total absolute weight carried by the ten highest-weight positions.

Mean-recovery error - $|\text{mean}(\text{data}[\text{sampled}]) - \text{mean}(\text{data})|$ on a numeric series, compared with a random sample of equal cardinality.

Time - wall-clock milliseconds for one call to the sampler (Python reference implementation).

The previously published "coverage efficiency" (0.67-0.70) is not used. It did not match any definition above and is withdrawn.

APPENDIX B - REPRODUCIBILITY

The core sampler is deterministic given fixed input. Seed 42 affects only synthetic data generation and stochastic baselines. Independent pure-Python re-implementation reproduces hit counts, weight sums, timing, and the mean-recovery result to within normal platform variation. Public-domain sources and the reference code are sufficient to regenerate every core number in this revision.

REFERENCES AND SOURCES

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